SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 3.4: Electrostatics — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.4: Electrostatics
Driving Question	DRIVING QUESTION: How do invisible electric charges build up, exert forces across empty space, and store energy — and how can we harness these abilities to protect communities and power technology in Kenya?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **What Is Happening Before the Lightning Strikes? (Introducing Charge)**	Students observed a balloon rubbed on hair/wool attracting small pieces of paper and sticking to the wall. When two rubbed balloons were brought close together, they repelled each other. A charged plastic pen held near a thin stream of water from a tap caused the stream to bend. Teacher note: Cold-call at least 3 students to describe what they saw before explaining why — surface-level observations only at this stage.	All matter is made of atoms containing positive protons (in the nucleus) and negative electrons (orbiting the nucleus). Objects become electrically charged when electrons are transferred from one material to another — the object gaining electrons becomes negatively charged; the one losing electrons becomes positively charged. The total charge in a closed system is always conserved (Law of Conservation of Charge). Materials fall into two groups: conductors (e.g., copper wire, iron roofing sheets common in Kenyan homes) allow electrons to move freely; insulators (e.g., rubber, dry wood, plastic) hold electrons tightly in place. Teacher note: Explicitly link the iron roof of a Kenyan home to the anchoring phenomenon — the roof is a conductor, so charges induced by a nearby storm cloud can redistribute across it rapidly.	When a cloud builds up excess negative charge at its base, it repels electrons in the iron roof directly below, leaving the roof's surface facing the cloud positively charged. This is the very first step toward a lightning strike. Students should be able to explain this using the atomic model: electron transfer occurred between water droplets and ice crystals colliding inside the cloud, giving the cloud's base a net negative charge. This connects directly to the driving question — 'invisible' charges have now been made visible through their effects (attraction, repulsion, bending water). Teacher note: Award full marks only if students explicitly name electron transfer (not just 'static electricity') and correctly identify which object gained/lost electrons.
Lesson 2 **Two Types of Charge, the Law of Charges, and Conductors vs.**	Students observed two ebonite rods rubbed with fur repelling each other, and an ebonite rod and a glass rod (rubbed with silk) attracting each other. A charged rod held	There are exactly two types of electric charge — positive and negative. The Law of Charges states: like charges repel; unlike charges attract. Charging by friction (also	The storm cloud's base, negatively charged by electron transfer between ice crystals and water droplets, exerts a repulsive force on electrons in the iron roof below (like

Insulators	near an electroscope caused the leaves to diverge. Students also tested household materials (plastic ruler, copper wire, rubber band, metal spoon) for conductivity using a simple circuit with a bulb. Teacher note: Ensure students record which materials lit the bulb (conductors) and which did not (insulators) in their notebooks before discussion — wait 2 minutes for independent recording.	called triboelectric charging) always transfers electrons — never protons, because protons are locked in the nucleus. The object with higher electron affinity gains electrons and becomes negative. Conductors (metals like copper, aluminium, iron) have free/delocalised electrons; insulators (plastic, rubber, glass, dry wood) do not. Teacher note: A common misconception is that positive objects 'gain positive charge' — stress repeatedly that only electrons move; positive charge arises from electron deficit.	charges repel) and an attractive force on the roof's positive ions (unlike charges attract). Because iron is a conductor with free electrons, this redistribution happens almost instantly across the entire roof. A rubber-thatched roof (insulator) would not redistribute charge this way — making the choice of roofing material relevant to lightning risk. This explains why metallic-roofed buildings in Nairobi, Kisumu, and rural Kenya require proper earthing. Teacher note: Use this contrast (metal roof vs. thatch) to connect Law of Charges directly to community safety — a key thread of the driving question.
Lesson 3 **Charging by Friction, Induction, and the Electrophorus — How Do Charges Separate Without Rubbing?**	Students observed a charged polythene strip brought near (but not touching) a neutral aluminium can on a non-conducting surface — the can rolled toward the strip. They then watched a teacher demonstration of an electrophorus: a charged plastic plate used repeatedly to charge a metal disc without losing its own charge. A Van de Graaff generator (or improvised version using a rotating belt) was demonstrated building up large amounts of charge. Teacher note: If a Van de Graaff is unavailable, a charged balloon near a neutral soda can on a smooth desk serves as an excellent substitute. Cold-call 2 students: 'Which end of the can is positive, and why?'	Charging by induction occurs when a charged object is brought near (not touching) a conductor, causing a redistribution of free electrons within the conductor — the near side becomes oppositely charged, the far side like-charged. If the conductor is then earthed (grounded) while the charged object is still near, electrons flow to/from earth, and when the earth connection is removed first (before the charged object), the conductor retains a net charge opposite to the inducing charge. Charging by friction generates charge through electron transfer when two different materials are rubbed (triboelectric series). The electrophorus uses induction + conduction repeatedly to accumulate charge without depleting the source. Teacher note: The four-step induction sequence (bring near → earth → remove earth → remove charged object) is a frequent exam question; drill it explicitly.	Induction explains how the iron roof of a Kenyan home can become strongly charged even though the storm cloud never touches it. As the negatively charged cloud base approaches, electrons in the roof are repelled downward/outward; the upper surface of the roof becomes positive. If the roof is connected to earth (as it should be via a lightning conductor), excess electrons flow away to earth, leaving the roof with a net positive charge. When lightning finally strikes, the discharge travels through the least resistant path — ideally a properly installed lightning conductor into the ground. Teacher note: Students often confuse the order of steps in induction; use a colour-coded diagram on the board (red = positive, blue = negative) and have students reproduce it in their notes.
Lesson 4 **Charging by Contact, Induction, and Earthing — How Did the Iron Roof Gain Charge Without Touching the Cloud?**	Students observed that touching a charged rod to a neutral electroscope caused the leaves to diverge and remain diverged after the rod was removed (charging by contact/conduction). In contrast, bringing a charged rod near — then earthing the electroscope, removing the earth connection, then removing the rod — left the leaves diverged with opposite charge	Charging by conduction (contact): the charged object touches the conductor, electrons transfer directly, and both objects end up with the same sign of charge. Charging by induction (no contact): the final charge on the conductor is opposite to the inducing charge (four steps must be followed in order). Earthing (grounding) connects a conductor to the Earth, which	A properly installed lightning conductor on a building in, say, Eldoret or Nakuru has a sharp metal spike at the top. Because charge concentrates at points, the spike develops an extremely high charge density, which ionises the surrounding air and creates a conducting plasma channel. Lightning preferentially follows this low-resistance path into the ground rather than

	(induction). A pointed conductor and a smooth conductor were compared using a flame or candle near each: the flame near the pointed conductor flickered more dramatically, illustrating the 'electric wind' from charge concentration at points. Teacher note: Allow students 3 minutes to compare and contrast the two charging methods in a T-chart before class discussion.	acts as an infinite reservoir of charge — electrons flow in or out until the conductor is neutral. Charge distributes unevenly on conductors: it concentrates at sharp points and edges, producing very high electric field strength there. This is the operating principle of the lightning rod (lightning conductor). Teacher note: Emphasise that a lightning rod does NOT prevent lightning — it provides a safe, low-resistance path to earth, protecting the structure. This is a common misconception to address explicitly.	passing through the building's structure. The iron roof's role is now fully explained: as a conductor, it redistributes charge by induction when the charged cloud approaches; if connected via a lightning conductor to earth, the charge is safely dissipated. Teacher note: This lesson is the conceptual centrepiece connecting charge behaviour to community protection — a core theme of the driving question. Ensure students can draw and label a complete lightning protection system.
Lesson 5 **How Strong Is the Push or Pull Between Charges? — Coulomb's Law**	Students observed a simulation (or physical demonstration using charged pith balls on threads) in which the angle of deflection of hanging charged balls increased as the balls were brought closer together or given more charge. Data was collected: force vs. distance and force vs. charge magnitude. Students plotted F vs. $1/r^2$ and F vs. q_1q_2, obtaining straight lines through the origin. Teacher note: If using a PhET simulation (Coulomb's Law), allow 5 minutes of free exploration before structured data collection. Cold-call 2 students to read values off their graphs before the class discusses the pattern.	Coulomb's Law: the electrostatic force between two point charges is directly proportional to the product of their charges and inversely proportional to the square of the distance between them: $F = kq_1q_2/r^2$, where $k = 9 \times 10^9$ N m² C⁻². The force is a vector — it acts along the line joining the two charges, repulsive if same sign, attractive if opposite sign. The unit of charge is the Coulomb (C); a single electron carries -1.6×10^{-19} C. Coulomb's Law is an inverse-square law, like Newton's Law of Gravitation, but electrostatic forces can be vastly stronger than gravitational forces for the same masses/charges. Teacher note: Walk students through a worked example set in Kenya — e.g., calculating the force between two charged raindrops in a Nairobi storm cloud separated by 1 cm, each carrying 2×10^{-9} C.	The enormous electrostatic force calculated using Coulomb's Law between the billions of separated charges in a storm cloud explains why lightning carries so much energy. Even though individual electron charges are tiny (10^{-19} C), when enormous numbers accumulate (total charges of coulombs), the forces and stored energy become destructive. Doubling the distance between the cloud base and the roof reduces the force by a factor of four — which is why taller buildings and trees are struck more frequently. This explains why farmers sheltering under tall trees during storms near Kericho or the Rift Valley are at significantly greater risk. Teacher note: Require students to show full substitution and unit analysis in all Coulomb's Law calculations; deduct marks for missing units or incorrect sign conventions.
Lesson 6 **Electric Fields — Mapping the Invisible Force Around Charges**	Students observed grass seeds or iron filings suspended in oil between charged plates and around point charges — the seeds aligned to reveal the shape of the electric field. Field line patterns were observed for: a single positive charge (lines radiating outward), a single negative charge (lines converging inward), two opposite charges (lines curving from + to −), two like charges (lines repelling, neutral point between them), and uniform field between parallel plates (straight parallel lines). Teacher note: If physical apparatus is	An electric field is a region of space in which a charged object experiences an electrostatic force. Electric field strength $E = F/q$ (force per unit positive test charge), measured in N C⁻¹ or V m⁻¹. For a point charge: $E = kQ/r^2$. Electric field lines show direction (from + to −, i.e., direction a positive test charge would move), never cross, and are denser where the field is stronger. Between two parallel plates carrying equal and opposite charge, the field is uniform (same strength and direction everywhere between the plates). The	The electric field of the storm cloud's negatively charged base extends downward through the air toward the iron roof below — this is the invisible 'action at a distance' the driving question asks about. The field is strongest near the sharp tip of a lightning conductor (highest charge density → highest field strength → $E = kQ/r^2$ is largest at smallest r). When the field strength exceeds approximately 3×10^6 V m⁻¹ (the breakdown strength of air), air molecules ionise and a conducting plasma channel forms — the leader stroke of lightning.

	unavailable, a PhET 'Charges and Fields' simulation works excellently. Give students 4 minutes to sketch observed field patterns before labelling them.	concept of field explains how charges exert forces across empty space — they create a field that permeates the surrounding space, and other charges in that field experience a force. Teacher note: Students frequently draw field lines crossing each other — stress that this is physically impossible (it would imply two directions of force at one point).	Mapping this field also explains the design of electrostatic precipitators used in Kenyan cement factories (e.g., in Athi River) to remove dust particles from emissions. Teacher note: Connect field line density explicitly to the Coulomb's Law result from Lesson 5 — stronger force = stronger field = denser field lines.
Lesson 7 **How Does Charge Store Energy — and How Can That Energy Be Released or Harnessed?**	Students observed a capacitor (two parallel aluminium foil plates separated by a plastic sheet) being charged by a battery, then discharged through a small LED — the LED briefly flashed. Energy stored was compared for different plate separations and areas. A demonstration showed a camera flash capacitor storing energy slowly (over seconds) and releasing it almost instantaneously. Students also observed that bringing the plates closer together (reducing r) caused a brighter flash, consistent with increased stored energy. Teacher note: Allow 2 minutes for students to record predictions about plate separation vs. brightness before the demonstration — this activates prior knowledge from Coulomb's Law ($F \propto 1/r^2$).	Work done separating opposite charges is stored as electric potential energy (EPE). Electric potential V at a point = work done per unit positive charge brought from infinity to that point; unit: volt ($V = J\ C^{-1}$). Potential difference (voltage) between two points = work done per unit charge moving between them: $W = QV$. A capacitor stores charge and energy: capacitance $C = Q/V$, unit farad (F). Energy stored in a capacitor: $E = ½CV^2 = ½QV = Q^2/2C$. Larger plate area and smaller plate separation increase capacitance. The electric potential energy stored in separated storm charges is ultimately converted to light, heat, sound, and radio waves during a lightning discharge. Teacher note: The equation $W = QV$ is the bridge between electrostatics and electrical circuits — emphasise this connection explicitly as it will reappear in the Electricity sub-strand.	The storm cloud acts as a giant natural capacitor: the negatively charged cloud base and the positively charged ground (or roof) below form two 'plates' separated by air. As charge builds up (Q increases) and the cloud descends (r decreases, increasing C), the stored energy $E = Q^2/2C$ increases enormously. When the air 'breaks down' and becomes conducting, all this stored energy is released in milliseconds — producing the blinding flash, the thunderclap, and temperatures exceeding 30,000 K in the lightning channel. A lightning conductor safely channels this energy to earth. Engineers in Kenya designing surge protection for M-Pesa mobile infrastructure and KPLC power lines use capacitor-based systems to absorb and dissipate such sudden energy releases. Teacher note: Students should be able to calculate stored energy given C and V, and explain qualitatively why energy release is so violent (very large Q, very small discharge time → enormous power = E/t).
Lesson 8 **Electrostatic Applications — How Do We Harness Charge to Protect and Power Kenya?**	Students examined labelled diagrams and short video clips (or teacher-led descriptions) of five electrostatic applications: (1) lightning conductors on tall buildings in Nairobi's CBD; (2) electrostatic precipitators at the Athi River cement factory removing dust from exhaust gases; (3) electrostatic spray painting used in vehicle manufacturing (e.g., at Kenya's vehicle assembly plants); (4) photocopiers and laser printers using charged drums to transfer toner; (5) Van de Graaff generators used in research and medical radiation	All five applications exploit the same core electrostatic principles covered in Lessons 1–7. Lightning conductors: sharp point → charge concentration → high field → air ionisation → safe discharge path to earth (induction + earthing + field concentration). Electrostatic precipitators: charged wire creates field that charges dust particles → charged particles attracted to oppositely charged collection plates → clean air released (charging by field/induction + Coulomb attraction). Spray painting: atomised paint droplets charged by	Each application in Lesson 8 is a direct answer to the driving question: humans have learned to control and harness the same physics that makes lightning dangerous. The precipitator answers 'protect communities' (cleaner air near factories). The lightning conductor answers 'protect communities' (safe discharge). The spray painter and photocopier answer 'power technology.' Engineers in Kenya working on the Nairobi expressway or standard gauge railway must consider electrostatic protection for control systems.

	therapy. Students identified which charging principle (friction, induction, conduction, field effects) underpins each application. Teacher note: Use a gallery-walk format if possible — post each application on a different wall; students rotate in groups of 3–4 and annotate sticky notes with the relevant physics principle.	friction/induction → attracted uniformly to grounded (earthed) metal object → even, efficient coat (reduces paint waste by up to 40%). Photocopier: light discharges selected areas of a charged selenium drum → toner (oppositely charged) sticks only to charged (dark) areas → transferred to paper by contact. Van de Graaff: continuous mechanical charge separation → accumulates very high voltage for research. Teacher note: Precipitators are particularly relevant to Kenya's industrialisation context — link to air quality in Mombasa (port emissions) and Athi River (cement/manufacturing).	A high-quality student response identifies at least three applications, names the specific charging mechanism involved, and connects it back to a principle from Lessons 1–7. Teacher note: This lesson is excellent for formative assessment — a well-answered exit ticket ('Choose one application, draw a labelled diagram, and explain the physics in 3 sentences') reveals whether students can transfer knowledge to new contexts.
Lesson 9 **Unit Review and Synthesis — Connecting Charge, Force, Field, Energy, and Application**	Students revisited the anchoring phenomenon — a photograph or video of lightning striking near a building in Nairobi — and the Driving Question Board accumulated over all 9 lessons. They observed that questions posted in Lesson 1 ('Why does lightning prefer tall objects?', 'Why does a metal roof feel dangerous?', 'How does a lightning rod work?') can now all be answered using unit vocabulary and equations. Students also observed a cumulative model they had built across lessons, showing the progression from atomic structure → electron transfer → Coulomb forces → electric fields → potential energy → applications. Teacher note: Give students 5 minutes of silent individual review of their DQB and model before the class synthesis discussion. Cold-call at least 4 students to answer archived DQB questions using full scientific language.	The complete electrostatics story: (1) Atoms contain protons (+) and electrons (−); electron transfer creates net charge (Lesson 1). (2) Like charges repel, unlike charges attract; conductors allow free electron flow, insulators do not (Lesson 2). (3) Charging methods — friction, conduction, induction — all conserve total charge (Lessons 2–4). (4) Charge concentrates at points on conductors; this underpins lightning rod design (Lesson 4). (5) Coulomb's Law ($F = kq_1q_2/r^2$) quantifies electrostatic force — an inverse-square relationship (Lesson 5). (6) Electric fields ($E = F/q = kQ/r^2$) map the force per unit charge in space; field lines show direction and strength (Lesson 6). (7) Separated charges store electric potential energy; capacitors store and release this energy (Lesson 7). (8) All these principles are harnessed in lightning conductors, precipitators, spray painters, photocopiers, and Van de Graaff generators (Lesson 8). Teacher note: The summary table itself — completed across all 9 lessons — is the primary artefact of this review lesson. Ensure every student has all 9 rows completed before the end of the lesson.	A complete, exemplary student explanation of the driving question reads as follows: 'Invisible electric charges build up in storm clouds when ice crystals and water droplets collide and transfer electrons — the cloud base becomes negatively charged. This charge exerts a Coulomb force ($F = kq_1q_2/r^2$) on charges in the iron roof below, and creates an electric field ($E = kQ/r^2$) that extends through the air. Because the roof is a conductor, its free electrons redistribute by induction, making the roof surface below the cloud positively charged. The separated charges store electric potential energy ($E = \frac{1}{2}QV$). When the field strength exceeds 3×10^6 V m^{-1}, air ionises and a plasma channel forms — lightning. A lightning conductor, with its sharp tip concentrating charge and a thick copper cable connecting to an earthing rod buried in moist soil, provides a low-resistance path for this energy to discharge safely into the ground, protecting homes and schools across Kenya. The same principles are used in precipitators to clean factory air in Athi River, in spray painting at vehicle plants, and in the photocopiers used in every Kenyan school.' Teacher note: Use this model response as a read-aloud exemplar and have students self-assess their own written responses against it using a 3-point rubric: (1) correct physics terms used, (2) correct equations

			cited, (3) Kenyan context connected.